

Rigidity of Cartesian C^1 degree-one axisymmetric self-similar Euler profiles with nonzero equatorial swirl amplitude

Wooseok Sung*

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Abstract

We study Cartesian C^1 axisymmetric steady self-similar profiles of the three-dimensional incompressible Euler equations that are homogeneous of degree one. In the exact-conical representation

$$\psi = r^3 f(z/r), \quad \Gamma = r^2 g(z/r),$$

we assume nonzero equatorial swirl amplitude,

$$B := g(0) \neq 0.$$

Under global Cartesian C^1 regularity, this condition forces the explicit linear strain plus rigid rotation about the symmetry axis

$$U_*(y) = \left(-\frac{y_1}{2} - By_2, By_1 - \frac{y_2}{2}, y_3\right), \quad B = g(0) \in \mathbb{R} \setminus \{0\},$$

with pressure $P_*(y) = (\frac{1}{8} + \frac{B^2}{2})(y_1^2 + y_2^2) - y_3^2$. The proof is a finite-dimensional linear-algebra consequence of C^1 degree-one homogeneity, axisymmetry, incompressibility, and the pressure compatibility condition. A Frobenius calculation also shows how zero-equatorial-amplitude conical branches, such as $g(\zeta) \sim D\zeta^2$, arise in the local ODE but are excluded by global Cartesian C^1 degree-one regularity.

1 Introduction

We consider the three-dimensional incompressible Euler equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = 0, \quad \operatorname{div} u = 0, \quad \text{on } \mathbb{R}^3 \times (0, T_*). \quad (1)$$

The usual self-similar substitution

$$u(x, t) = (T_* - t)^{\gamma-1} U((T_* - t)^{-\gamma} x) \quad (2)$$

leads to the steady profile equation

$$(1 - \gamma)U + \gamma(y \cdot \nabla)U + (U \cdot \nabla)U + \nabla P = 0, \quad \operatorname{div} U = 0. \quad (3)$$

Liouville-type results for this equation, and for the stationary Leray system in Navier–Stokes, usually impose decay, integrability, finite-energy, or boundary-geometry assumptions on the profile. Examples include Chae’s Euler criteria [2, 1], the Luo–Hou cylinder analysis of Chae–Tsai [3, 6], the Leray-profile theorem of Nečas–Růžička–Šverák [7], Tsai’s local-energy theorem and pointwise Liouville lemma [8], and recent constraints on putative self-similarity for incompressible 3D Euler by Constantin–Ignatova–Vicol [4].

*School of Computer Science, College of Electrical and Computer Engineering, Chungbuk National University.
Email: tjddntjr1@chungbuk.ac.kr.

Here we isolate globally Cartesian C^1 , axisymmetric, degree-one homogeneous solutions of (3). The explicit family

$$U_*(y) = \left(-\frac{y_1}{2} - By_2, By_1 - \frac{y_2}{2}, y_3\right), \quad B \in \mathbb{R} \setminus \{0\}, \quad (4)$$

is smooth and divergence-free on all of $\mathbb{R}^3$. It has constant vorticity $\text{curl } U_* = 2Be_3$, grows linearly in y , and solves the steady self-similar profile equation exactly, for every $\gamma \in \mathbb{R}$, with

$$P_*(y) = \left(\frac{1}{8} + \frac{B^2}{2}\right)(y_1^2 + y_2^2) - y_3^2.$$

For $B \neq 0$, the ordinary steady Euler residual

$$(U_* \cdot \nabla)U_* + \nabla P_*$$

is nonzero; the profile solves the self-similar equation only after the linear term in (3) is included.

For an axisymmetric field written in cylindrical coordinates, we use the stream function and circulation

$$u^r = -\frac{\psi_z}{r}, \quad u^z = \frac{\psi_r}{r}, \quad u^\theta = \frac{\Gamma}{r}.$$

In the degree-one exact-conical class these take the form

$$\psi = r^3 f(\zeta), \quad \Gamma = r^2 g(\zeta), \quad \zeta = \frac{z}{r}. \quad (5)$$

The swirl hypothesis is the equatorial amplitude condition

$$B := g(0) \neq 0.$$

This condition has different force in the global and local settings. In the globally Cartesian C^1 degree-one class, homogeneity will force the conical swirl profile to be constant. In the local analytic ODE calculation, by contrast, a nonzero analytic function g may vanish at $\zeta = 0$; replacing the displayed assumption by $g \not\equiv 0$ permits additional local branches.

Main theorem. In the globally Cartesian C^1 , degree-one homogeneous, axisymmetric self-similar Euler profile class, nonzero equatorial swirl amplitude $g(0) \neq 0$ forces the explicit field (4).

The proof starts from the Cartesian C^1 degree-one assumption. A C^1 degree-one homogeneous vector field on all of $\mathbb{R}^3$ is linear. Axisymmetry and incompressibility give a two-parameter matrix family, and the pressure compatibility condition forces the coefficient $1/2$ in the presence of nonzero equatorial swirl. A Frobenius calculation then shows the same forcing in conical ODE variables and identifies the differentiability-at-the-origin obstruction for zero-equatorial-amplitude branches.

The self-similar exponent plays no role in this degree-one class: Euler's identity gives $(y \cdot \nabla)U = U$, so $(1 - \gamma)U + \gamma(y \cdot \nabla)U = U$ for every $\gamma \in \mathbb{R}$.

Classical Liouville theorems provide the function-space comparison. The profile (4) has constant nonzero vorticity and linear velocity growth; the results listed above require decay or integrability. In particular, Tsai's pointwise Liouville lemma assumes the sublinear condition $V(y) = o(|y|)$.

2 Conical formulation and main theorem

We use the axisymmetric notation and state the main degree-one rigidity theorem. The theorem concerns globally Cartesian C^1 velocity fields; local conical ODE branches outside that global class are discussed in §3.4.

2.1 Cylindrical Euler in (ψ, Γ) form

We work with the steady self-similar profile equation

$$(1 - \gamma)U + \gamma(y \cdot \nabla)U + (U \cdot \nabla)U + \nabla P = 0, \quad \operatorname{div} U = 0, \quad y \in \mathbb{R}^3. \quad (6)$$

Here $\gamma \in \mathbb{R}$ is the self-similar exponent. For axisymmetric fields, we use the following convention. Let R_ϕ be rotation by angle ϕ about the y_3 -axis. We call a vector field U axisymmetric if

$$U(R_\phi y) = R_\phi U(y)$$

for every ϕ . Swirl is allowed.

For axisymmetric fields in cylindrical coordinates (r, θ, z) , write $U = u^r e_r + u^\theta e_\theta + u^z e_z$ and introduce the stream function ψ and the circulation $\Gamma := ru^\theta$ by

$$u^r = -\frac{\psi_z}{r}, \quad u^z = \frac{\psi_r}{r}, \quad u^\theta = \frac{\Gamma}{r}. \quad (7)$$

This representation builds in incompressibility. Let

$$\mathcal{L}\psi := \psi_{rr} - \frac{1}{r}\psi_r + \psi_{zz}.$$

Then $\omega^\theta = -\mathcal{L}\psi/r$ and $\eta := \omega^\theta/r = -\mathcal{L}\psi/r^2$.

Eliminating the pressure by taking ∂_z of the radial component of (6) minus ∂_r of the axial component, then dividing by r , gives

$$(1 + \gamma)\eta + \gamma S\eta + D\eta - \frac{1}{r^4}(\Gamma^2)_z = 0, \quad (8)$$

where $S := r\partial_r + z\partial_z$ and $D := u^r\partial_r + u^z\partial_z$. Multiplying the θ -component of (6) by r gives

$$(1 - 2\gamma)\Gamma + \gamma S\Gamma + D\Gamma = 0. \quad (9)$$

2.2 Degree-one conical normalization

In the degree-one class studied below, the conical powers are fixed by the velocity degrees:

$$\psi(r, z) = r^3 f(\zeta), \quad \Gamma(r, z) = r^2 g(\zeta), \quad \zeta := \frac{z}{r}. \quad (10)$$

Indeed, $u^r = -\psi_z/r$ and $u^z = \psi_r/r$ have degree one precisely when ψ has degree three, and $u^\theta = \Gamma/r$ has degree one precisely when Γ has degree two. The argument below starts from this degree-one normalization. Arbitrary conical powers would require separate non-degeneracy assumptions.

Substituting (10) into (7) gives

$$u^r = -rf'(\zeta), \quad u^z = r(3f(\zeta) - \zeta f'(\zeta)), \quad u^\theta = rg(\zeta). \quad (11)$$

The velocity is homogeneous of degree one on $\{r > 0\}$.

For a degree-one axisymmetric field written on $\{r > 0\}$ as

$$u^\theta(r, z) = rg(z/r),$$

we call $g(0)$ the equatorial swirl amplitude.

For later comparison with the reduced ODE, set

$$h(\zeta) := (1 + \zeta^2)f''(\zeta) - 3\zeta f'(\zeta) + 3f(\zeta). \quad (12)$$

With the normalization (10), one has

$$\mathcal{L}\psi = rh(\zeta), \quad \omega^\theta = -h(\zeta), \quad \eta = -\frac{h(\zeta)}{r}.$$

The reduced vorticity equation (8) becomes

$$3fh' + (1 + f')h + 2gg' = 0, \quad (13)$$

and the reduced circulation equation (9) becomes

$$3fg' - 2f'g + g = 0. \quad (14)$$

Equations (12)–(13)–(14) provide the conical check in §3.3.

Remark 2.1 (Equatorial swirl amplitude). Analyticity permits a nonzero function g to vanish at $\zeta = 0$, so the amplitude assumption is stated directly. In the globally Cartesian C^1 degree-one class, homogeneity forces the velocity to be linear, and the conical swirl profile is constant. In that class, $g(0) = 0$ implies $g \equiv 0$.

2.3 Main theorem

Theorem 2.2 (Rigidity with nonzero equatorial swirl amplitude). *Let $U \in C^1(\mathbb{R}^3; \mathbb{R}^3)$ be axisymmetric, divergence-free, and homogeneous of degree one:*

$$U(\lambda y) = \lambda U(y), \quad \lambda > 0.$$

Suppose that U solves (6) on $\mathbb{R}^3$ for some $\gamma \in \mathbb{R}$ and some smooth scalar pressure P . In the exact-conical representation (10), assume

$$B := g(0) \neq 0.$$

Then

$$U(y) = \left(-\frac{y_1}{2} - By_2, By_1 - \frac{y_2}{2}, y_3\right), \quad B \in \mathbb{R} \setminus \{0\}. \quad (15)$$

Up to an additive constant,

$$P(y) = \left(\frac{1}{8} + \frac{B^2}{2}\right)(y_1^2 + y_2^2) - y_3^2. \quad (16)$$

The vorticity is the constant axial field $\text{curl } U = 2Be_3$.

The proof is given in §3. It uses the global Cartesian C^1 regularity and degree-one homogeneity before entering the conical ODE variables.

Remark 2.3 (Scope). The theorem concerns degree-one fields. Modulated ansatzes such as $\psi = r^{3-\delta}f(\zeta)$, $\Gamma = r^{2-\delta}g(\zeta)$ with $\delta \neq 0$, logarithmic corrections $\psi = r^3(\log r)^\beta f(\zeta)$, or stretched variables $\zeta = z/r^{1+\epsilon}$ lead to a different problem: explicit residual dependence on r replaces the autonomous system (13)–(14).

3 Linear rigidity and conical expansions

Global Cartesian C^1 regularity enters before the reduced ODEs. A C^1 degree-one homogeneous field on all of $\mathbb{R}^3$ is linear. Axisymmetry and incompressibility leave a two-parameter matrix family, and the pressure condition forces the coefficient once the equatorial swirl amplitude is nonzero. The Frobenius calculation below is a local check in the variable $\zeta = z/r$.

3.1 Degree-one C^1 homogeneity

Lemma 3.1 (Degree-one C^1 homogeneity implies linearity). *Let $U \in C^1(\mathbb{R}^3; \mathbb{R}^3)$ satisfy*

$$U(\lambda y) = \lambda U(y), \quad \lambda > 0.$$

Then $U(y) = DU(0)y$ for every $y \in \mathbb{R}^3$.

Proof. Homogeneity gives $U(0) = 0$. For fixed y and $t > 0$,

$$\frac{U(ty) - U(0)}{t} = U(y).$$

Letting $t \downarrow 0$ in the difference quotient gives $U(y) = DU(0)y$. □

3.2 Proof of the main theorem

Proof of Theorem 2.2. By Lemma 3.1, $U(y) = My$ for a real 3×3 matrix M . Axisymmetry means that M commutes with every rotation about the y_3 -axis. There are $A, b, C \in \mathbb{R}$ such that

$$M = \begin{pmatrix} -A & -b & 0 \\ b & -A & 0 \\ 0 & 0 & C \end{pmatrix}.$$

Since $\operatorname{div} U = \operatorname{tr} M = 0$, we have $C = 2A$.

For this matrix field,

$$u^\theta = br, \quad \Gamma = br^2.$$

Thus in the conical representation $g(\zeta) \equiv b$, and the equatorial swirl amplitude is $g(0) = b$.

For a degree-one linear field, $(y \cdot \nabla)U = U$. The self-similar equation (6) reduces to

$$U + (U \cdot \nabla)U + \nabla P = 0.$$

With $U = My$, this is

$$(M + M^2)y + \nabla P = 0.$$

A vector field of the form $y \mapsto Ny$ is a gradient exactly when N is symmetric, so $M + M^2$ must be symmetric. Direct computation gives

$$M + M^2 = \begin{pmatrix} A^2 - A - b^2 & b(2A - 1) & 0 \\ b(1 - 2A) & A^2 - A - b^2 & 0 \\ 0 & 0 & 2A(2A + 1) \end{pmatrix}.$$

Symmetry is equivalent to

$$b(1 - 2A) = 0.$$

Since $g(0) = b = B \neq 0$, we get $A = 1/2$. This gives the velocity in (15). Integrating $\nabla P = -(M + M^2)y$ gives (16), up to an additive constant. □

For any differentiable degree-one field, Euler's identity gives $(y \cdot \nabla)U = U$, so the profile equation is independent of γ in this class.

Remark 3.2 (Swirl-free contrast). If $B = 0$, the condition $b(1 - 2A) = 0$ imposes no restriction on A . Nonzero equatorial swirl amplitude is the hypothesis that forces $A = 1/2$ inside the degree-one axisymmetric linear family.

3.3 Reduction on the nonzero-amplitude branch

The reduced equations from §2 are

$$3fh' + (1 + f')h + 2gg' = 0, \quad (17)$$

$$3fg' - 2f'g + g = 0, \quad (18)$$

where

$$h = (1 + \zeta^2)f'' - 3\zeta f' + 3f.$$

As a local calculation in ζ , the expansion shows how the Cartesian C^1 linear branch appears in the conical variables; the global Cartesian C^1 hypothesis is a separate condition.

Assume that f and g are analytic near $\zeta = 0$, that $g(0) = B \neq 0$, and that we are in the equatorially reflection-symmetric parity sector

$$f(-\zeta) = -f(\zeta), \quad g(-\zeta) = g(\zeta).$$

Thus

$$f(\zeta) = A\zeta + C\zeta^3 + O(\zeta^5), \quad g(\zeta) = B + D\zeta^2 + O(\zeta^4). \quad (19)$$

Proposition 3.3 (Frobenius reduction on the nonzero-amplitude branch). *Under the assumptions above, any such analytic parity-sector solution of (17)–(18) agrees, in a possibly smaller neighborhood of $\zeta = 0$, with*

$$f(\zeta) = \frac{\zeta}{2}, \quad g(\zeta) = B.$$

Proof. The constant term of (18) is $B(1 - 2A)$. Since $B \neq 0$, $A = 1/2$.

At order ζ^2 in (18), one obtains $D = 2BC$. With $A = 1/2$,

$$h(\zeta) = 6C\zeta + O(\zeta^3), \quad h'(\zeta) = 6C + O(\zeta^2).$$

The coefficient of ζ in (17) is

$$18C + 4BD.$$

Substituting $D = 2BC$ gives

$$(18 + 8B^2)C = 0. \quad (20)$$

The coefficient $18 + 8B^2$ is positive, so $C = 0$ and then $D = 0$.

The same calculation repeats at higher orders. Suppose that all earlier coefficients have vanished and write, for $k \geq 1$,

$$f(\zeta) = \frac{\zeta}{2} + E_k\zeta^{2k+1} + O(\zeta^{2k+3}), \quad g(\zeta) = B + F_k\zeta^{2k} + O(\zeta^{2k+2}).$$

The swirl equation at order ζ^{2k} gives

$$F_k = \frac{2(2k+1)}{3k}BE_k,$$

and the vorticity equation at order ζ^{2k-1} gives

$$E_k \left[6k^2(2k+1) + \frac{8(2k+1)}{3}B^2 \right] = 0.$$

The bracket is positive for every $k \geq 1$ and $B \in \mathbb{R}$, so $E_k = 0$ and then $F_k = 0$. Analyticity gives the asserted equality in a neighborhood of $\zeta = 0$. $\square$

Appendix A.3 gives the coefficient calculation in closed form.

3.4 Zero-equatorial branches

Without the amplitude condition $g(0) \neq 0$, the local ODE admits vanishing-equator balances. If

$$f(\zeta) = A\zeta + \cdots, \quad g(\zeta) = D\zeta^m + \cdots, \quad D \neq 0, \quad m \geq 1,$$

then the leading part of the swirl equation

$$3fg' - 2f'g + g = 0$$

is

$$D(A(3m-2) + 1)\zeta^m = 0.$$

Thus

$$A = -\frac{1}{3m-2}.$$

For reflection-even branches $m = 2\ell$, $\ell \geq 1$, this becomes

$$A = -\frac{1}{6\ell-2}.$$

In particular, $g(\zeta) \sim D\zeta^2$ gives $A = -1/4$.

These local branches are incompatible with globally Cartesian C^1 exact degree-one fields on $\mathbb{R}^3$. The decisive obstruction is differentiability at the origin. If a degree-one homogeneous field is C^1 at the origin, Lemma 3.1 gives

$$U(y) = DU(0)y.$$

In the axisymmetric linear class the swirl component has the form

$$u^\theta = br,$$

so the conical swirl profile is constant:

$$g(\zeta) \equiv b.$$

Thus any local conical branch with

$$g(\zeta) = D\zeta^m + \cdots, \quad m \geq 1, \quad D \neq 0,$$

cannot belong to the globally Cartesian C^1 exact degree-one class. The conical ODE is detecting a local germ outside the regularity class of Theorem 2.2.

4 Comparison with Liouville-type hypotheses

Classical Liouville theorems place the explicit profile in its function-space context. Theorem 2.2 classifies one Cartesian C^1 degree-one class; the cited theorems impose decay or integrability hypotheses absent for U_* . The comparison below is only meant to locate the profile relative to those hypotheses.

For U_* ,

$$\operatorname{curl} U_* = 2Be_3,$$

so the vorticity has infinite $L^p(\mathbb{R}^3)$ norm for every finite p . Chae's vorticity-integrability criterion for self-similar Euler profiles [2] is unavailable for U_* . The same constant vorticity is also incompatible with the limiting-vorticity integrability condition in Chae's asymptotic theorem [1].

The velocity has linear growth:

$$|U_*(y)| = \Theta(|y|).$$

The L^3 norm required by Nečas–Růžička–Šverák [7] is infinite, as are the finite- q norms in Tsai [8]. For $q = \infty$, Tsai’s hypothesis requires a bounded profile; U_* is unbounded.

Tsai’s pointwise Liouville lemma is the closest comparison. In that notation, the velocity assumption is $U(y) = o(|y|)$ at infinity, with polynomial pressure control. The profile U_* has polynomial pressure and linear velocity growth, while the lemma assumes sublinear velocity growth.

Chae–Tsai [3] treat a Luo–Hou type ansatz on the cylinder $\{0 < r < 1\} \times \mathbb{R}$, with boundary and decay assumptions in rescaled variables [6]. Here the domain is all of $\mathbb{R}^3$, and the inputs are global Cartesian C^1 regularity plus exact degree-one homogeneity.

Thus the present theorem does not strengthen the decay-based Liouville results. It classifies a separate exact degree-one axisymmetric class, where nonzero equatorial swirl amplitude forces the explicit linear strain plus rigid rotation about the symmetry axis.

5 Discussion

The argument uses four structural inputs: global Cartesian C^1 regularity, exact degree-one homogeneity, axisymmetry, and nonzero equatorial swirl amplitude. Their roles are as follows.

Regularity at the origin. The step from homogeneity to linearity uses differentiability at the origin. If differentiability at the origin is dropped, degree-one homogeneity need not force linearity. This issue is separate from Elgindi’s $C^{1,\alpha}$ construction [5], which concerns a different dynamic, regularity, and profile class.

Axisymmetry. Without axisymmetry, the degree-one C^1 problem becomes a finite matrix problem: classify trace-free matrices M for which $M + M^2$ is symmetric, modulo the natural action of rotations. The axisymmetric theorem is the two-parameter subcase in which this condition reduces to $b(1 - 2A) = 0$, with b the linear swirl coefficient.

The swirl amplitude. If the equatorial swirl amplitude is zero, the matrix compatibility condition imposes no constraint on the strain parameter A in the axisymmetric family. In conical variables, local branches with $g(0) = 0$ but $g \not\equiv 0$ arise before global Cartesian C^1 degree-one regularity is imposed; §3.4 gives the leading balance and the obstruction to Cartesian C^1 regularity. Accordingly, the geometric hypothesis is $g(0) \neq 0$.

Exact degree-one homogeneity. Modulated conical ansatzes such as $\psi = r^{3-\delta}f(\zeta)$, $\Gamma = r^{2-\delta}g(\zeta)$, logarithmic corrections, or stretched variables lead away from the autonomous degree-one matrix problem. Their analysis is closer to a residual problem for a perturbed profile equation, as in recent numerical profile searches [9].

The result separates local conical ODE branches from globally Cartesian C^1 degree-one Euler profiles. In that axisymmetric degree-one class, nonzero equatorial swirl amplitude forces the explicit linear strain plus rigid rotation about the symmetry axis. Classical decay-based Liouville theorems apply in different function classes.

A Verification of U_* and the Frobenius recursion

A.1 Explicit form of U_*

For the profile in Theorem 2.2, the exact-conical ansatz (10) of Section 2 has $\psi = r^3 \cdot \zeta/2 = r^2 z/2$ and $\Gamma = r^2 \cdot B = Br^2$. The stream-function recovery (7),

$$\begin{aligned} u^r &= -\psi_z/r = -(r^2/2)/r = -r/2, \\ u^z &= \psi_r/r = (2rz/2)/r = z, \\ u^\theta &= \Gamma/r = Br. \end{aligned}$$

In Cartesian coordinates, use

$$(y_1, y_2, y_3) = (r \cos \theta, r \sin \theta, z), \quad e_r = (\cos \theta, \sin \theta, 0),$$

and

$$e_\theta = (-\sin \theta, \cos \theta, 0), \quad e_z = (0, 0, 1).$$

Then

$$\begin{aligned} U_*(y) &= u^r e_r + u^\theta e_\theta + u^z e_z \\ &= \left(-\frac{r}{2} \cos \theta - Br \sin \theta, -\frac{r}{2} \sin \theta + Br \cos \theta, z\right) \\ &= \left(-\frac{y_1}{2} - By_2, By_1 - \frac{y_2}{2}, y_3\right). \end{aligned} \tag{21}$$

A.2 Verification of the steady self-similar profile equation

We verify that U_* given by (21) solves the steady self-similar Euler system on $\mathbb{R}^3$ with the explicit pressure

$$P_*(y) = \left(\frac{1}{8} + \frac{B^2}{2}\right)(y_1^2 + y_2^2) - y_3^2. \tag{22}$$

Incompressibility. $\partial_1 U_*^1 + \partial_2 U_*^2 + \partial_3 U_*^3 = -\frac{1}{2} - \frac{1}{2} + 1 = 0$.

Vorticity. The Jacobian of U_* is the constant matrix

$$\nabla U_* = \begin{pmatrix} -\frac{1}{2} & -B & 0 \\ B & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

so $\text{curl } U_* = (0, 0, B - (-B)) = 2B e_3$. The vorticity is the constant axial field $\Omega_* = 2B e_3$.

Steady self-similar Euler. For the degree-one homogeneous profile U_* , Euler's identity gives $y \cdot \nabla U_* = U_*$ (direct check on the Jacobian: $(y \cdot \nabla)U_*^1 = -\frac{y_1}{2} - By_2 = U_*^1$, and analogously for the other components). The self-similar Euler equation (6) with parameter γ reduces, by the γ -cancellation $(1 - \gamma)U_* + \gamma y \cdot \nabla U_* = U_*$, to

$$U_* + (U_* \cdot \nabla)U_* + \nabla P_* = 0, \tag{23}$$

independent of γ . Component-wise,

$$\begin{aligned} [(U_* \cdot \nabla)U_*]_1 &= U_*^j \partial_j U_*^1 = \left(\frac{1}{4} - B^2\right)y_1 + By_2, \\ [(U_* \cdot \nabla)U_*]_2 &= -By_1 + \left(\frac{1}{4} - B^2\right)y_2, \\ [(U_* \cdot \nabla)U_*]_3 &= y_3. \end{aligned}$$

Therefore

$$U_* + (U_* \cdot \nabla)U_* = \left(-\left(\frac{1}{4} + B^2\right)y_1, -\left(\frac{1}{4} + B^2\right)y_2, 2y_3\right).$$

With P_* as in (22), $\nabla P_* = \left(\left(\frac{1}{4} + B^2\right)y_1, \left(\frac{1}{4} + B^2\right)y_2, -2y_3\right)$, so $U_* + (U_* \cdot \nabla)U_* + \nabla P_* = 0$ identically on $\mathbb{R}^3$. The pressure P_* is unique up to additive constant.

A.3 Higher-order recursion details

This subsection gives the explicit form of the inductive recursion step used in the Frobenius calculation of Section 3.3 to show that all Taylor coefficients of f and g beyond $A = 1/2$ and $g(0) = B$ vanish.

Suppose by induction that, for some $k \geq 1$,

$$f(\zeta) = \frac{\zeta}{2} + E_k \zeta^{2k+1} + O(\zeta^{2k+3}), \quad g(\zeta) = B + F_k \zeta^{2k} + O(\zeta^{2k+2}), \quad (24)$$

i.e. all earlier nontrivial Taylor coefficients of f and g have been forced to vanish.

The swirl equation at order ζ^{2k} . From $f' = \frac{1}{2} + (2k+1)E_k \zeta^{2k} + O(\zeta^{2k+2})$ and $g' = 2kF_k \zeta^{2k-1}$, the swirl ODE (14) at order ζ^{2k} gives $3kF_k - 2(2k+1)BE_k = 0$, giving

$$F_k = \frac{2(2k+1)}{3k} B E_k. \quad (25)$$

The vorticity equation at the matched order. The leading ζ^{2k-1} contribution to $3fh' + (1+f')h + 2gg' = 0$ takes the form

$$\alpha_k E_k + \beta_k B F_k = 0 \quad (26)$$

for explicit constants $\alpha_k, \beta_k > 0$ depending only on k . Substituting (25) into (26) gives

$$E_k \cdot p_k(B^2) = 0, \quad p_k(B^2) := \alpha_k + \beta_k \cdot \frac{2(2k+1)}{3k} B^2. \quad (27)$$

Lemma A.1 (Closed form of the recursion coefficients). *For every $k \geq 1$,*

$$\alpha_k = 6k^2(2k+1), \quad \beta_k = 4k, \quad (28)$$

and consequently

$$p_k(B^2) = 6k^2(2k+1) + \frac{8(2k+1)}{3} B^2 > 0 \quad \text{for all } k \geq 1, B \in \mathbb{R}. \quad (29)$$

In particular $E_k = 0$ and (by (25)) $F_k = 0$ for every $k \geq 1$.

Proof. Substitute (24) into $h(\zeta) := (1+\zeta^2)f''(\zeta) - 3\zeta f'(\zeta) + 3f(\zeta)$ at order ζ^{2k-1} . From $f = \frac{\zeta}{2} + E_k \zeta^{2k+1} + O(\zeta^{2k+3})$ we have $f' = \frac{1}{2} + (2k+1)E_k \zeta^{2k} + O(\zeta^{2k+2})$ and $f'' = 2k(2k+1)E_k \zeta^{2k-1} + O(\zeta^{2k+1})$. The linear contributions from $-3\zeta f'$ and $3f$ at order ζ^1 cancel exactly ($-3\zeta \cdot \frac{1}{2} + 3 \cdot \frac{\zeta}{2} = 0$), so the leading term of h at order ζ^{2k-1} comes purely from $(1+\zeta^2)f''$:

$$h(\zeta) = 2k(2k+1)E_k \zeta^{2k-1} + O(\zeta^{2k+1}), \quad h'(\zeta) = 2k(2k+1)(2k-1)E_k \zeta^{2k-2} + O(\zeta^{2k}). \quad (30)$$

Collecting the ζ^{2k-1} coefficients of $3fh' + (1+f')h + 2gg'$ using (30) and $g' = 2kF_k \zeta^{2k-1}$:

$$\begin{aligned} [3fh']_{\zeta^{2k-1}} &= 3 \cdot \frac{\zeta}{2} \cdot 2k(2k+1)(2k-1)E_k \zeta^{2k-2} = 3k(2k+1)(2k-1)E_k, \\ [(1+f')h]_{\zeta^{2k-1}} &= \frac{3}{2} \cdot 2k(2k+1)E_k = 3k(2k+1)E_k, \\ [2gg']_{\zeta^{2k-1}} &= 2B \cdot 2kF_k = 4kB F_k. \end{aligned}$$

The first two sum to $3k(2k+1)[(2k-1)+1]E_k = 6k^2(2k+1)E_k$, giving $\alpha_k = 6k^2(2k+1)$ and $\beta_k = 4k$. Substituting (25) into (26) yields $E_k \cdot [6k^2(2k+1) + \frac{8(2k+1)}{3}B^2] = 0$. Both summands in the bracket are positive for $k \geq 1$ and any real B , so $E_k = 0$ and then $F_k = 0$ by (25). $\square$

As a check, the $k = 1$ case of (28) gives $(\alpha_1, \beta_1) = (18, 4)$ and $p_1(B^2) = 18 + 8B^2$, recovering equation (20) of the main text with $C \rightarrow E_1$, $D \rightarrow F_1$. The $k = 2$ case gives $(\alpha_2, \beta_2) = (120, 8)$, $p_2(B^2) = 120 + \frac{40}{3}B^2$.

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